

Zerui Chen | 陈泽瑞

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Available: 30 Jun 2027 - 30 Sep 2027

Education

University of Cambridge | BA (Hons) Computer Science | *Cambridge, UK* **Expected Jul 2027**

Related Coursework: Artificial Intelligence, Advanced Data Science, Bayesian Inference, Deep Neural Networks, Natural Language Processing, Geometric Deep Learning, Reinforcement Learning (expected by graduation)

Hwa Chong Institution | GCE Advanced Levels | *Singapore* **Jan 2016 - Dec 2021**

- Physics, Computing, Mathematics, Knowledge & Inquiry: **AAAA/AA**;
- Semiconductor Physics & Devices [NTU]: **Distinction**
- Participated and obtained **Champion** in numerous **hackathons** and **data science competitions**:
 - Build on Singapore 2020, DSTA Cyberthon 2021, DSTA Brainhack 2021, NUS Medical Grand Challenge 2024.

Professional Experience

WorldQuant | Quantitative Research Intern | *Singapore* **06 Jul 2026 - 30 Sep 2026**

ByteDance, Global E-Commerce | Machine Learning Engineer Intern | *Singapore* **30 Jun 2025 - 30 Sep 2025**

- Independently researched and developed novel and robust **auxiliary objectives** and **model architectural** optimizations for TikTok Global Shop's fine-ranking DL model, trained on in-house **HPC** clusters, achieving **+0.3% revenue lift**.
- Built internal tools for interactive model architecture **visualization** and automated **training behavior analysis**.

[Competition] NUS Medical Grand Challenge | Full-Stack & LLM Developer | *Global* **Feb 2024 - Aug 2024**

- Founded medical-AI startup *Tomy* to revolutionize remote stoma care emerged **Grand Champion** amongst 40 teams from universities across Asia, partnering with **5 Singapore public hospitals** with deployed clinical trials.
- Implemented and deployed image classification API using few-shot prompted vision language model, achieving **100% recall** and **80% precision** in 4-fold cross-validation with extremely scarce training data.

Singapore Armed Forces | Signals & Admin Specialist | *Singapore* **Jan 2022 - Nov 2023**

- Developed duty assignment algorithm with availability constraints, saving hours monthly and improving fairness.
- Led training/exercise planning for signals platoon, achieving 95% top-grade readiness.

Agency for Science, Technology and Research | Bio-AI Research Intern | *Singapore* **Jun 2020 - May 2021**

- Developed a LSTM-CNN deep learning model for atrial tachycardia diagnosis, reaching SOTA-comparable 81% accuracy while being highly data-efficient; parallelized data ETL pipeline to achieve **100× throughput**.
- Co-authored and published a comparative study on decision trees vs deep learning models: [arxiv/2206.07515](#).

Technical Skills

- **Machine Learning** with [Python](#) using PyTorch, TensorFlow; expertise in deep learning for NLP, CV, RecSys.
- **Data ETL & Analysis** with [SQL](#), MongoDB, NumPy, Pandas, Matplotlib;
- **iOS/Web Development** with [Objective-C](#), [Swift 6](#), UIKit, SwiftUI; [JS ES6/Typescript 5](#), SCSS, Vue, React
- **General Development** with [C++](#), [Java](#), [OCaml](#), [Rust](#)
- **Languages:** English (native), Mandarin Chinese (native).

Selected Projects

- **[nanodiscover]** A clean reimplementation of [TTT-Discover](#), a **SOTA test-time-learning automated discovery system** (online RL over a quality-diversity archive). Adopted as the **test-time-RL scaffold** in Evolution Fine-Tuning (co-authored; [AID-Wild Workshop @ ACM CAIS 2026](#)). **Ongoing research** building on it to systematically analyze ADS **failure modes** and propose **targeted interventions** for search efficiency & effectiveness, deployed on HPC clusters.
- **[Github zlab]** A **modular deep learning research platform** for rapid experimentation with transformer architectures, enabling **systematic ablation studies** across model architectures and training configurations via a registry-driven composition system with built-in experiment management for grid search and metric-tracked evaluation.
- **[Github MML-Book]** Independently typesetting the full solutions to the [Mathematics for Machine Learning book](#), implementing with basic NumPy ops the core **linear algebra**, **regression** and **bayesian inference** algorithms.

Interests

Exploring the topic of **robust**, **reliable** and **explainable inference** with artificial intelligence, and ultimately its application to efficiently model other systems.